

Requirements Specification Table

Airport Lost Luggage Claim & Tracking Portal (Problem Statement #30)

PES University - Dept. of CSE | SE Lab 1

Target System: Airline Baggage Recovery Platform

Target Stakeholders / Actors: Passenger, Baggage Service Agent

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	Cross-reference passenger lost baggage reports against unrouted luggage scan logs across all connected airport terminals.	High	Pass: Match identified & recovery alert generated. Fail: Incorrect bag tag format accepted.	Enables automated matching of misplaced luggage across transit hubs.
FR-002	Functional	Allow passengers to submit a digital lost baggage claim using Flight Booking Reference (PNR) and bag tag number.	High	Pass: Claim ID generated with status 'Submitted'. Fail: Claim submitted without mandatory bag tag number.	Eliminates manual queueing at physical service desks during baggage loss.
FR-003	Functional	Display a real-time scan timeline showing multi-leg baggage transit locations and handling statuses.	Medium	Pass: Multi-leg scans display correct timestamp & airport code. Fail: Timeline omits intermediate transit scan.	Provides transparent tracking to passengers and reduces support inquiry volume.
FR-004	Functional	Allow Baggage Service Agents to manually override and update baggage scan status and transit locations.	High	Pass: Status updates reflect in portal within 5 seconds. Fail: Unauthorized agent overrides status log.	Empowers ground staff to update routing manually during offline exceptions.
FR-005	Functional	Automatically initiate compensation calculation workflows for baggage delayed beyond 24 hours.	Medium	Pass: Compensation claim opened automatically at 24h mark. Fail: Delayed bag requires manual system trigger.	Ensures compliance with automated passenger compensation protocols.
NFR-001	Performance & Security	The baggage tracking timeline API must return complete multi-leg scan history in under 200 ms under peak load.	High	Pass: Benchmarking tests confirm target latency (<200 ms). Fail: Latency exceeds target standard.	Ensures high system responsiveness during peak travel disruption events.
NFR-002	Security & Privacy	All passenger claim records and Personally Identifiable Information (PII) must be encrypted at rest (AES-256) and in transit (TLS 1.3).	High	Pass: Security audit confirms active end-to-end encryption. Fail: Unencrypted PII detected in network log.	Protects sensitive passenger personal data against breaches and regulatory non-compliance.